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PDF 15

Data Science and AI Glossary

143 terms and acronyms in plain English, each with an example and related terms.

Part of the AI for Data Engineers (Data Science & AI) programme.

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Data Science and AI Glossary

143 terms and acronyms with a simple definition, a real example and related terms, in A to Z order.

A

Term	Meaning	Example	Related
A/B test	An experiment that splits users into two groups to compare a change with the current version.	Half of users see a new checkout button; compare conversion.	Hypothesis test, p-value, sample size
ACID Atomicity, Consistency, Isolation, Durability	Guarantees that database transactions are reliable.	A bank transfer either fully happens or not at all.	Transaction, OLTP
Agent	An AI system that plans and uses tools in a loop to reach a goal.	An agent searches orders, drafts a reply and asks for approval.	Tool use, orchestration, MCP
Aggregation	Combining many rows into a summary such as a sum, count or average.	Total sales per region.	GROUP BY, pivot table
Airflow	An open-source tool for scheduling and monitoring data pipelines as DAGs.	A nightly DAG loads orders and builds a revenue table.	DAG, orchestration, ETL
Anomaly detection	Finding data points that are unusually different from the rest.	Flagging a sudden spike in refunds.	Outlier, z-score, monitoring
API Application Programming Interface	A defined way for one program to request data or actions from another.	A payment app calls the bank's API to confirm a payment.	Endpoint, REST, webhook
AUC Area Under the Curve	A single number summarising a ROC or precision-recall curve.	ROC-AUC of 0.85 means good ranking of positives.	ROC curve, classification
Avro	A row-based file format with an attached schema, common in streaming.	Kafka messages stored as Avro with a schema registry.	Parquet, schema, Kafka

B

Term	Meaning	Example	Related
Batch processing	Processing data in scheduled chunks rather than continuously.	Loading yesterday's orders at 2 a.m.	Streaming, ETL
BI Business Intelligence	Tools and practices that turn data into dashboards and reports for decisions.	A Power BI sales dashboard.	Dashboard, KPI, data warehouse
Bias (model)	Systematic error, including unfair differences in outcomes between groups.	A loan model that rejects one region far more often for the same risk.	Fairness, responsible AI
Big O notation	A way to describe how running time or memory grows with input size.	Searching a sorted list with binary search is O(log n).	DSA, complexity
Bootstrap	Estimating uncertainty by resampling the data with replacement many times.	A confidence interval for median order value.	Confidence interval, sampling

C

Term	Meaning	Example	Related

CDC Change Data Capture	Copying only the rows that changed in a source system.	Streaming new and updated orders to the warehouse.	Incremental load, ETL
Central limit theorem CLT	Averages of large samples are roughly normally distributed, whatever the original distribution.	Why confidence intervals for averages work.	Sampling, normal distribution
Chunking	Splitting documents into smaller passages for retrieval.	One help-article section per chunk.	RAG, embedding
Classification	Predicting a category.	Will this customer churn: yes or no?	Regression, precision, recall
Clustering	Grouping similar items without labels.	Customer segments from purchase behaviour.	K-means, unsupervised learning
Cohort analysis	Comparing groups that started at the same time.	Retention of users who signed up in January vs February.	Retention, churn
Confidence interval CI	A range likely to contain the true value, given the data.	Conversion 3.0% ± 0.4 points (95% CI).	Standard error, p-value
Confusion matrix	A table of correct and incorrect predictions by class.	TP 32, FP 18, FN 8, TN 142.	Precision, recall
Context engineering	Designing everything an AI sees when it answers: instructions, documents, data, memory and tools.	A support assistant gets the order record and policy, not the whole database.	Prompt engineering, RAG, context window
Context window	The maximum number of tokens a model can consider in one request.	Instructions, documents, chat and answer must fit.	Token, LLM
Correlation	How strongly two variables move together, from −1 to 1.	Ad spend and sales with r = 0.7.	Causation, regression
Cross-validation	Training and testing a model on several data splits for a stable score.	5-fold cross-validation of a churn model.	Overfitting, validation set
CSV Comma-Separated Values	A plain-text table with one row per line.	sales.csv exported from Excel.	Parquet, JSON
CTE Common Table Expression	A named temporary result inside a SQL query, written with WITH.	WITH monthly AS (...) SELECT ...	Subquery, SQL

D

Term	Meaning	Example	Related
DAG Directed Acyclic Graph	A set of steps with dependencies and no loops, used to define pipelines.	Extract, then transform, then load.	Airflow, orchestration
Dashboard	A page of charts and numbers that tracks key metrics.	Daily orders, revenue and delivery time.	BI, KPI
Data catalog	A searchable inventory of datasets with owners and definitions.	Finding which table holds 'active customers'.	Metadata, governance, lineage
Data contract	An agreement on a dataset's schema, meaning and quality between producer and consumer.	The app team promises order_id is never null.	Schema, data quality

Data governance	Rules, roles and processes for using data correctly and safely.	Who may access salary data, and why.	Stewardship, privacy, lineage
Data lake	Cheap storage holding raw data in many formats.	Raw logs and files in cloud object storage.	Lakehouse, data warehouse
Data lineage	The record of where data came from and how it changed.	Revenue table built from orders and refunds.	Catalog, governance
Data mart	A subject-focused part of a warehouse for one team.	A finance mart with revenue and cost tables.	Data warehouse, star schema
Data observability	Monitoring data freshness, volume, schema and quality to catch problems early.	Alert when today's row count drops by half.	Data quality, monitoring
Data pipeline	Automated steps that move and transform data.	App database to warehouse to dashboard.	ETL, ELT, orchestration
Data quality	How accurate, complete, consistent, timely and valid data is.	No duplicate order IDs; prices above zero.	Validation, observability
Data warehouse	A database designed for analysis, holding cleaned, modelled data.	Snowflake, BigQuery, Redshift, Fabric warehouse.	Data lake, OLAP, star schema
DAX Data Analysis Expressions	The formula language of Power BI and Analysis Services.	Total Sales = SUM(Sales[Amount])	Measure, filter context
dbt data build tool	A tool that turns SQL SELECT statements into tested, documented warehouse tables.	A model builds fact_orders with a not-null test.	ELT, data warehouse
Decision tree	A model that splits data by simple yes/no questions.	Is tenure under 6 months? Then high churn risk.	Random forest, gradient boosting
Deep learning	Machine learning with many-layered neural networks.	Image recognition, speech, LLMs.	Neural network, transformer
Dimension table	A table of descriptive attributes used to slice facts.	Product, Customer, Date.	Fact table, star schema
Docker	A tool that packages an application and its dependencies into a container.	Running the same pipeline on a laptop and a server.	Container, deployment

E

Term	Meaning	Example	Related
EDA Exploratory Data Analysis	First look at data to understand its shape, quality and patterns.	Histograms, missing-value counts, correlations.	Data cleaning, visualisation
ELT Extract, Load, Transform	Load raw data first, then transform it inside the warehouse.	Load orders raw, build models with dbt.	ETL, dbt
Embedding	A list of numbers that captures the meaning of text, images or other data.	'refund' and 'money back' have similar embeddings.	Vector database, RAG
Ensemble	Combining several models to improve accuracy.	Random forests average many decision trees.	Bagging, boosting

ETL Extract, Transform, Load	Extract data, transform it, then load it into the target.	Clean CSVs in Python before loading the warehouse.	ELT, pipeline
Eval Evaluation	A repeatable test of how well an AI system does a job, on fixed cases with clear scoring.	Pass rate on 200 support questions.	Golden set, LLM-as-judge

F

Term	Meaning	Example	Related
Fact table	A table of measurable events with keys to dimensions.	Sales rows with date, product, customer and amount.	Dimension table, star schema
Feature	An input variable used by a model.	Days since last order.	Feature engineering, label
Feature engineering	Creating useful model inputs from raw data.	Turning order dates into 'orders in last 30 days'.	Feature, preprocessing
Fine-tuning	Further training a pre-trained model on your own examples.	Teaching a model your support tone.	LoRA, RAG, prompting
Forecasting	Predicting future values from past data.	Next month's demand per store.	Time series, seasonality
Foreign key FK	A column that refers to another table's primary key.	orders.customer_id points to customers.id.	Primary key, join

G

Term	Meaning	Example	Related
Gradient boosting	Builds trees one after another, each fixing the previous errors.	XGBoost or LightGBM for churn.	Decision tree, ensemble
Gradient descent	An optimisation method that adjusts parameters step by step to reduce error.	Training a linear regression.	Learning rate, loss function
Guardrails	Checks on AI inputs and outputs that block unsafe or invalid results.	Reject replies containing phone numbers.	Safety, evals, prompt injection

H

Term	Meaning	Example	Related
Hallucination	An AI output that sounds confident but is not supported by facts or sources.	An invented refund policy.	Grounding, RAG, evals
Hyperparameter	A setting chosen before training rather than learned.	Tree depth, learning rate.	Tuning, cross-validation
Hypothesis test	A method to judge whether data shows a real effect.	Is the new checkout better?	p-value, A/B test

I

Term	Meaning	Example	Related

Idempotency	Repeating an operation has the same effect as doing it once.	A retried payment request does not charge twice.	API, retry
Imputation	Filling missing values with estimates.	Replace a missing age with the median.	Missing data, data cleaning
Incremental load	Loading only new or changed data since the last run.	Orders with updated_at after the last load.	CDC, batch

J

Term	Meaning	Example	Related
JSON JavaScript Object Notation	A text format for structured data used by most APIs.	{"id": 42, "status": "paid"}	API, schema

K

Term	Meaning	Example	Related
K-means	A clustering method that groups points around k centres.	Five customer segments.	Clustering, unsupervised
Kafka	A platform for streaming events between systems.	Order events consumed by analytics and email services.	Streaming, event
KPI Key Performance Indicator	A metric that shows progress toward a goal.	Monthly active users.	Metric, dashboard, OKR

L

Term	Meaning	Example	Related
Label	The answer a supervised model learns to predict.	Churned: yes or no.	Feature, supervised learning
Lakehouse	A platform combining lake storage with warehouse features such as tables and transactions.	Delta Lake tables in Databricks or Fabric.	Data lake, data warehouse
Linear regression	Predicts a number as a weighted sum of inputs.	House price from size and location.	Regression, R-squared
LLM Large Language Model	A model trained on large text collections that generates text by predicting tokens.	Claude, GPT, Gemini, Llama.	Token, transformer
LLM-as-judge	Using a language model to grade another model's outputs against a rubric.	A judge checks whether answers are faithful to sources.	Eval, rubric
Logistic regression	A classification model that outputs a probability.	Probability that a lead converts.	Classification, sigmoid

M

Term	Meaning	Example	Related

MAE Mean Absolute Error	The average size of prediction errors.	Off by ₹120 on average.	RMSE, regression
MCP Model Context Protocol	An open standard for connecting AI assistants to tools and data sources.	An MCP server gives an assistant read access to a database.	API, tool use, agent
Measure (DAX)	A calculation in Power BI that responds to filters.	Sales YTD.	DAX, filter context
Medallion architecture	Organising lake data into bronze (raw), silver (clean) and gold (business-ready) layers.	Gold table feeds the revenue dashboard.	Lakehouse, data quality
Metadata	Data about data: names, types, owners, update times.	Column descriptions in a catalog.	Data catalog, lineage
MLOps Machine Learning Operations	Practices for deploying, monitoring and updating models in production.	Retraining a churn model monthly with checks.	Model drift, monitoring
Model drift	Model quality drops because data or behaviour changes over time.	A pricing model trained before a festival season.	Monitoring, retraining

N

Term	Meaning	Example	Related
Normalisation (database)	Organising tables to reduce duplication.	Customers in one table, orders in another.	Primary key, foreign key
Normalisation (scaling)	Rescaling features to a common range.	Min-max scaling to 0–1.	Standardisation, preprocessing
NoSQL	Databases that store data as documents, key-values, wide columns or graphs instead of tables.	MongoDB for product catalogs.	SQL, schema
Null	A missing or unknown value.	A customer with no phone number.	Imputation, COALESCE

O

Term	Meaning	Example	Related
OAuth Open Authorization	A standard that lets a user grant an app limited access without sharing a password.	Sign in with Google, then allow calendar read access.	Authentication, API
OLAP Online Analytical Processing	Systems designed for analysis over large data.	Warehouse queries for monthly revenue.	OLTP, data warehouse
OLTP Online Transaction Processing	Systems designed for many small, fast transactions.	The app database that stores each order.	OLAP, ACID
One-hot encoding	Turning a category into separate 0/1 columns.	City = Pune becomes is_pune = 1.	Feature engineering
Orchestration	Coordinating the order, timing and retries of steps in a pipeline or AI workflow.	Airflow runs extract before transform.	DAG, workflow, agent

Outlier	A value far from the rest of the data.	A ₹5,00,000 order among ₹500 orders.	IQR, anomaly detection
Overfitting	A model learns noise in training data and fails on new data.	Train accuracy 99%, test 70%.	Underfitting, regularisation

P

Term	Meaning	Example	Related
p-value	The chance of seeing a result at least this extreme if there were no real effect.	p = 0.03 in an A/B test.	Hypothesis test, significance
Parquet	A columnar file format that is compressed and fast for analytics.	Lake tables stored as Parquet.	CSV, Avro, columnar
Partitioning	Splitting a large table into parts, often by date, so queries read less data.	Orders partitioned by order_date.	Performance, data warehouse
PCA Principal Component Analysis	Reduces many correlated features to a few summary components.	Compress 50 survey questions into 5 factors.	Dimensionality reduction
PII Personally Identifiable Information	Data that can identify a person.	Name, phone, email, Aadhaar number.	Privacy, masking, DPDP Act
Pivot table	A summary table that groups and aggregates data.	Sales by region and month.	Aggregation, Excel
Power Query	The data import and transformation tool in Excel and Power BI.	A repeatable cleaning step for monthly files.	ETL, Excel, Power BI
Precision	Of the items predicted positive, the share that were truly positive.	64% of flagged tickets were urgent.	Recall, F1
Primary key PK	A column that uniquely identifies each row.	order_id.	Foreign key, normalisation
Prompt engineering	Writing instructions that get reliable results from a language model.	Role, goal, format, examples and rules.	Context engineering, system prompt
Prompt injection	Text in an AI's input that tries to override its instructions.	'Ignore your rules' hidden in an email.	Guardrails, security

R

Term	Meaning	Example	Related
R-squared R^2	The share of variation in the target that a regression explains.	R^2 = 0.72.	Regression, RMSE
RAG Retrieval-Augmented Generation	Searching your documents and giving the relevant passages to an LLM to answer from.	A help-centre assistant that cites articles.	Embedding, vector database, chunking
Random forest	An ensemble of decision trees trained on random samples.	Predicting churn with 300 trees.	Decision tree, ensemble
Recall	Of the truly positive items, the share the model found.	80% of urgent tickets were caught.	Precision, F1

Recommendation system	A model that suggests items a user is likely to want.	'Customers also bought'.	Collaborative filtering
Regression	Predicting a number.	Next month's revenue.	Classification, MAE, RMSE
Regularisation	Penalising model complexity to reduce overfitting.	L1 (lasso) and L2 (ridge).	Overfitting
REST Representational State Transfer	A common style for web APIs using URLs and HTTP methods.	GET /orders/42.	API, endpoint
Retention	The share of users who keep using a product over time.	40% of January users active in March.	Churn, cohort
RLS Row-Level Security	Restricting which rows a user can see.	Regional managers see only their region in Power BI.	Security, governance
RMSE Root Mean Squared Error	The square root of the average squared error; larger errors count more.	RMSE of ₹180.	MAE, regression
ROC curve Receiver Operating Characteristic	A plot of true positive rate against false positive rate across thresholds.	Used to compare classifiers.	AUC, threshold

S

Term	Meaning	Example	Related
Sampling	Selecting part of a population to study.	Surveying 1,000 of 50,000 customers.	Bias, confidence interval
SCD Slowly Changing Dimension	A way to track changes to dimension attributes over time.	Type 2 keeps old and new customer addresses with dates.	Dimension table, data warehouse
Schema	The structure of data: tables, columns, types and rules.	orders(id INT, amount DECIMAL).	Data contract, DDL
Seasonality	Patterns that repeat on a calendar cycle.	Higher sales every Diwali.	Time series, forecasting
Spark	A distributed engine for processing large data across many machines.	Aggregating billions of events with PySpark.	Big data, Databricks
SQL Structured Query Language	The standard language for querying and changing relational databases.	SELECT city, SUM(amount) FROM orders GROUP BY city;	Database, join
Standard deviation SD	How spread out values are around the mean.	Delivery time 30 ± 6 minutes.	Variance, z-score
Star schema	A data model with a central fact table linked to dimension tables.	Sales fact with Date, Product, Customer.	Fact table, dimension table
Statistical significance	A result unlikely to be due to chance alone, judged by a threshold.	$p < 0.05$.	p-value, A/B test
Streaming	Processing events continuously as they arrive.	Fraud checks on each card swipe.	Kafka, batch
Supervised learning	Learning from examples that have known answers.	Churn model trained on past churn labels.	Label, classification, regression

T

Term	Meaning	Example	Related
Temperature	A setting that controls how random a language model's output is.	Low for extraction, higher for brainstorming.	LLM, sampling
Test set	Data held back to measure final model performance.	20% of rows never used in training.	Validation set, overfitting
Time series	Data recorded in time order.	Daily orders for two years.	Forecasting, seasonality
Token	A unit of text a language model reads or writes, roughly ¾ of an English word.	Prices are quoted per million tokens.	Context window, LLM
Tool use Function calling	A model asks your code to run a defined function and uses the result.	get_order_status(order_id).	Agent, MCP
Transformer	The neural network design behind modern LLMs, based on attention.	GPT, Claude and Gemini are transformers.	LLM, attention

U

Term	Meaning	Example	Related
Underfitting	A model too simple to capture the pattern.	A straight line through curved data.	Overfitting
Unsupervised learning	Finding structure in data without labels.	Customer clustering.	Clustering, PCA

V

Term	Meaning	Example	Related
Validation set	Data used to tune a model before the final test.	Choosing tree depth on validation scores.	Test set, cross-validation
Vector database	A database that stores embeddings and finds the most similar ones quickly.	pgvector, Pinecone, Chroma.	Embedding, RAG

W

Term	Meaning	Example	Related
Webhook	An HTTP call an app sends to your URL when an event happens.	Payment succeeded posts to your server.	API, event
Window function	A SQL function that calculates across related rows without collapsing them.	ROW_NUMBER() OVER (PARTITION BY city ORDER BY amount DESC).	SQL, ranking

X

Term	Meaning	Example	Related
XLOOKUP	An Excel function that looks up a value and returns a matching result.	=XLOOKUP(G2, A:A, C:C).	INDEX MATCH, VLOOKUP

Z

Term	Meaning	Example	Related
z-score	How many standard deviations a value is from the mean.	z = 2.5 suggests an unusual value.	Standard deviation, outlier